

Instruction:

For the homework, a dataset will be provided to you on Canvas, and you need to download it for use. A final Jupyter Notebook file (.ipynb) needs to be submitted as your answers to the homework questions.

You are strongly encouraged to work on this problem set together in the form of a study group. However, everyone should submit his/her individual write-up.

Questions:

You are provided with the raster data from Cropland Data Layer (CDL) in 2022 and 2023 for California, US. Additionally, there is a csv file about the land types, which is a dictionary. A gpkg file of California Counties is provided for drawing the shape of California and the counties. Use this in all plots.

1. (10 points) Read the two raster files and plot them.
2. (15 points) Summarize the top 10 land uses (by area) in California in each year. Rank from top 1 to top 10. [You need to use the dictionary here. Ignore when Codes=0, which is just background]
3. (20 points) Create two rasters for each year:
 - 1 if pixel is almonds, 0 otherwise
 - 1 if pixel is grapes, 0 otherwiseAnd then plot them. [Check the dictionary to have the Codes representing almonds and grapes. When plotting, you can downsample for speed.]
4. (10 points) Describe the degree of crop rotation (almonds and grapes only) between 2022 and 2023. [Use count of pixels.]
5. (15 points) Create a raster using the CDL layers where each pixel is labeled:
 - 1 = almonds stayed almonds
 - 2 = grapes stayed grapes
 - 3 = almonds changed to grapes
 - 4 = grapes changed to almonds
 - 0 = any other case
6. (10 points) Visualize the raster data you created in question 5. Label the legend clearly. It's a good practice to ignore "0 = any other case" to make the graph background look cleaner.
7. (10 points) Suppose a client has the following question:

"What are the crop yield advantages of crop rotation as compared to monocropping in California?"

Describe how you might answer this question using the CDL layer, specifically: - What data sources you would need in addition to the raster you just created. - How you would manipulate the data. - What statistics you would calculate or model you would run on this data. - How you might present your findings. [This is not a coding question.]

8. (10 points) What factors would make it difficult to accurately answer the question? List three and explain why they might make answering the question difficult. [This is not a coding question.]